IIT Guwahati - Department of Computer Science & Engineering

CS 2201 – Computer Architecture & Organization

Mid - Semester Examination - (16.09.2026: 120 minutes, 60 marks)

1. *Index the facing sheet with question numbers and corresponding page numbers.*
2. *Show calculations and steps wherever necessary.*
3. *Part A (Q1-Q8) must be written in main answer booklet and Part B (Q9 & Q10)must be written in the QP itself and handover Part B QP sheet separately along with main answer booklet. For Part B, write your name and roll number in the space provided and get it signed by invigilator.*

Q. No	Q:1	Q:2	Q:3	Q:4	Q:5	Q:6	Q:7	Q:8	Q:9	Q:10	Total
Marks	6	4	4	4	4	4	8	10	8	8	60

Part A

1. Answer the following.
   (a) Name the foreign company collaborating with HCL for setting up OSAT plant in Jewar, Uttar Pradesh.
   (b) Name the company that owns and manufactures the Ryzen processor series.
   (c) Name the world's largest contract chipmaker company from Taiwan.
   (d) Name the author of the book '*Chip War*'.
   (e) Name any one recipient of the 2025 ACM A.M. Turing Award.
   (f) What is the full form of SPEC? (benchmark suite)

   Ans: [1 mark each]

   (a)Foxconn (Hon Hai Technology Group)

   (b)AMD (Advanced Micro Devices, Inc.)

   (c)TSMC (Taiwan Semiconductor Manufacturing Company)

   (d)Chris Miller

   (e)Andrew G. Barto and Richard S. Sutton

   (f)SPEC = Standard Performance Evaluation Corporation

2. Match the following with unique mapping from Set A to Set B.

Set A	Set B
(a) RAW Hazard	(p) Benchmark suite
(b) WAR Hazard	(q) Control hazard
(c) Tomasulo's Algorithm	(r) Inorder commit
(d) Loop Unrolling	(s) Register renaming
(e) CoreMark	(t) CDB pinging
(f) FP-DIV unit	(u) Score boarding
(g) ROB	(v) Structural hazard
(h) Delay slot	(w) Register pressure

   Ans: [0.5 mark each]

   a → t

   b → s

   c → u

d → w

e → p

f → v

g → r

h → q

3. Assume that a machine A has an 8-stage pipeline with a 1 ns clock cycle. Another machine B has a 12-stage pipeline with a 0.9 ns clock cycle. A incurs 2 stalls due to a data hazard for every 7 instructions, whereas the B experiences 3 stalls for every 11 instructions. In addition, branch mis-prediction rate for both machines is 10%. Branches constitute 30% of the instructions. The branch mis-prediction penalty for A is 2 cycles but for B it is 5 cycles. What is the speedup of the B over A (round off to 2 decimal places)?

Machine A:

Clock cycle time = 1 ns

Stalls due to data hazard per instruction = $\frac{2}{7}$ stalls

Branch misprediction penalty = 2 cycles

Machine B:

Clock cycle time = 0.9 ns

Stalls due to data hazard per instruction = $\frac{3}{11}$ stalls

Branch misprediction penalty = 5 cycles

Branches = 30% = 0.3

Mis-prediction rate = 10% = 0.1

$CPI_{ideal} = 1$

$CPI = CPI_{ideal} + CPI_{data} + CPI_{branch}$

Machine A:

Branch mis-prediction per instruction = 0.3 x 0.1 = 0.03

Branch stall cycles per instruction = 0.03 x 2 = 0.06 cycles [0.5 mark]

$CPI_A = CPI_{ideal} + CPI_{data} + CPI_{branch}$

$CPI_A = 1 + \frac{2}{7} + 0.06 = 1 + 0.2857 + 0.06 = 1.3457$ [0.5 mark]

Time per instruction (T_A) = 1.3457 x 1 ns = 1.3457 ns [0.5 mark]

Machine B:

Branch mis-prediction per instruction = 0.3 x 0.1 = 0.03

Branch stall cycles per instruction = 0.03 x 5 = 0.15 cycles [0.5 mark]

$CPI_B = CPI_{ideal} + CPI_{data} + CPI_{branch}$

$CPI_B = 1 + \frac{3}{11} + 0.15 = 1 + 0.2727 + 0.15 = 1.4227$ [0.5 mark]

Time per instruction (T_B) = 1.4227 x 0.9 ns = 1.2804 ns [0.5 mark]

Speedup of B over A = $\frac{T_A}{T_B} = \frac{1.3457}{1.2804} = 1.05$ [1 mark]

Machine B is 1.05 times faster than Machine A.

4. A program consists of two categories of instructions: S-sequential and P-parallel. The execution time for each S-type as well as P-type instruction is 1 ms on a unicore processor. P-type instructions get speeded up by factor of N upon running on an N-core processor. The program which took 1800 ms in a unicore processor took only 300 ms upon running on a 16-core processor. Let *F1* and *F2* be the fraction of time spent on executing S-type instructions on a unicore processor and 16-core processor, respectively. What will be impact ratio of the sequential code, if we define the impact ratio as $\frac{F1}{F2}$?

Execution time on unicore processor, T_1 = 1800 ms

Execution time on 16-core processor, T_{16} = 300 ms

F_1 is the fraction of uniprocessor execution time spent on sequential instructions.

Execution time for sequential instructions on unicore processor, $T_s = 1800 \times F_1$

Execution time for parallel instructions on unicore processor, $T_P = 1800 \times (1 - F_1)$

On 16-core processor, sequential instructions don't speedup, but parallel instructions get a speedup of 16.

T_{16} = 300 ms

$T_s + \frac{T_p}{16} = 300$ ms

$1800 \times F_1 + \frac{1800 \times (1-F_1)}{16} = 300$

Solve for F_1; $\mathbf{F_1 = \frac{1}{9}}$ **[2 marks]**

$T_s = 1800 \times \frac{1}{9} = 200$ ms

F_2 is the fraction of 16-core processor execution time spent on sequential instructions.

$\mathbf{F_2 = \frac{T_s}{T_{16}} = \frac{200\ ms}{300\ ms} = \frac{2}{3}}$ **[1 mark]**

$\frac{F1}{F2} = \frac{1}{6}$ **[1 mark]**

5. A new floating-point unit speeds up floating-point operations by three times. In an application one fifth of the instructions are floating-point operations.

(a) What is the overall speedup? (Ignore the penalty to other instructions).

Fraction $_{enhanced}$ =⅕ =0.2

Speedup$_{enhanced}$ =3

$$Speedup_{overall} = \frac{1}{(1 - Fraction_{enhanced}) + \frac{Fraction_{enhanced}}{Speedup_{enhanced}}}$$

$\frac{1}{(1 - 0.2) + \frac{0.2}{3}} = \frac{1}{0.8 + 0.0667} = \frac{1}{0.8667} = 1.154$ [1 mark]

(b) Assume that the speeding up of the floating-point unit mentioned above slowed down data cache accesses resulting in a 2 times slowdown. Assuming the load instructions constitute 15% and store instructions constitute 9% of the total instruction what is the effective overall speedup now?

fx = 0.20 fy= 0.15 fz= 0.09

Nx = 3 Ny =½ = 0.5 Nz=½=0.5

$$S = \frac{1}{(1 - fx - fy - fz) + \left(\frac{fx}{Nx} + \frac{fy}{Ny} + \frac{fz}{Nz}\right)}$$

step 1: (1- fx - fy- fz) = 1- 0.20 - 0.15 - 0.09 =0.56

step 2: $\frac{fx}{Nx} + \frac{fy}{Ny} + \frac{fz}{Nz} = \frac{0.20}{3} + \frac{0.15}{0.5} + \frac{0.09}{0.5}$ **[2 marks]**

= 0.0667+0.30+0.18 = 0.5467

Step 3 : $S = \frac{1}{0.56+0.5467} = \frac{1}{1.1067}$

S= 0.9036 [1 mark]

6. Write down the instruction sequence to evaluate the expression **S=(B*B)+(A*C)** for the following architecture specifications. Here **A, B, C** and **S** are memory locations.

(a) Stack machine. **[1 marks]**

PUSH B
PUSH B
MUL
PUSH A
PUSH C
MUL
ADD
POP S

(b) Accumulator machine **[1.5 marks]**

LOAD B
MUL B
STORE T

```
LOAD A
MUL C
ADD T
STORE S
```

(c) Load Store machine with four registers R1, R2, R3 and R4 **[1.5 mark]**

```
LOAD R1,B
MUL R2,R1,R1
LOAD R3,A
LOAD R4,C
MUL R1,R3,R4
ADD R3,R2,R1
STORE S,R3
```

7. A RISC machine with latencies of Load, Store, Add.s (Add scalar) and Mult.s (Multiply scalar) are 1, 0, 2 and 3 respectively. Branch delay slot is one cycle. Consider a loop as given below. X[i] is a float array with 30 elements and A and B are scalar numbers.

```
for (int i = 29; i >= 0; i--) {
        X[i] = A.X[i] + B }
```

   (a) What is the equivalent RISC-MIPS code for this loop? (Hint: each element in the array has to be loaded from memory to an FP register, multiply it with A, add B to it and store it back in the same location. Assume that the value of A and B are already available in F0 and F1, respectively. Adjust memory index address and check exit condition to repeat the loop.) How many stalls are there in each iteration of the loop? What is the total number of stalls for executing the program?

   (b) Perform loop unrolling with three elements processed in one iteration for the code you generated in (a) above. How many stalls are there in each iteration of the loop? What is the total number of stalls for executing the program?

   (c) Use pipeline scheduling (show the code) on the unrolled loop you generated in (b) above. How many stalls are there in each iteration of the loop? What is the total number of stalls for executing the program?

Ans: (A)

```
LOOP:
    L.S     F2, 0(R1)
    # 1 stall
    MUL.S   F3, F2, F0
    # 3 stalls
    ADD.S   F4, F3, F1
    # 2 stalls
    S.S     F4, 0(R1)
```

```
DADDUI  R1, R1, #-4
# 1 stall
BNE     R1, R2, LOOP
```
[1 Mark]

For 1 iteration, 7 stalls.

Total = 7 x 30 = 210 stalls [1 Mark (each and total)]

(B)

Loop Unrolling

```
LOOP:
    L.S     F2, 0(R1)
    MUL.S   F3, F2, F0
    ADD.S   F4, F3, F1
    S.S     F4, 0(R1)
    L.S     F5, -4(R1)
    MUL.S   F6, F5, F0
    ADD.S   F7, F6, F1
    S.S     F7, -4(R1)
    L.S     F8, -8(R1)
    MUL.S   F9, F8, F0
    ADD.S   F10, F9, F1
    S.S     F10, -8(R1)
    DADDUI  R1, R1, #-12
    BNE     R1, R2, LOOP
```
[2 Marks]

For 1 iteration, 19 stalls.

Total = 19 x 10 = 190 stalls

(C)

Pipeline Scheduling on Unrolled Loop

```
LOOP:
    L.S     F2, 0(R1)
    L.S     F5, -4(R1)
    MUL.S   F3, F2, F0
    L.S     F8, -8(R1)
    MUL.S   F6, F5, F0
    MUL.S   F9, F8, F0
```

```
ADD.S   F4, F3, F1
DADDUI  R1, R1, #-12
ADD.S   F7, F6, F1
S.S     F4, 12(R1)
ADD.S   F10, F9, F1
S.S     F7, 8(R1)
BNE     R1, R2, LOOP
S.S     F10, 4(R1)
```

[2 Marks]

For 1 iteration, 0 stalls.

Total = 0 stalls [1 Mark (each and total)]

8. Mention whether each of the following statements given below is True/False. Explain with justifications.

(a) In a dynamically scheduled speculative processor, which has one reservation station per functional unit, if the register status indicator of a register R4 is 3, then the latest value of R4 can be obtained from the CDB when functional unit #3 writes a result.

False. In a speculative processor, the Register Status Indicator holds a ROB entry index (or tag), not a Functional Unit index. The value for R4 will be obtained when the instruction corresponding to ROB Entry #3 completes and broadcasts on the CDB, regardless of which physical functional unit executed it. (1+1)

(b) In a MIPS multi-cycle floating point pipeline that supports operand forwarding, there will be 7 stalls between a pair of adjacent FMUL instructions that has a RAW dependency between them.

False. Let I1 and I2 have RW dependency. Result of I1 available for forwarding at the end of 9th cycle. I2 starts execution in the 10th cycle. Number of stalls = 10-4 = 6 stalls. (1+1)

(c) A (p, q) type branch predictor uses the outcome of last 2^p branches to index into the branch handling table, where each entry has a q-bit predictor.

False. A (p, q) predictor uses the last p branch outcomes to index the branch-history table, with each entry containing a q-bit predictor. (1+1)

(d) In a dynamic scheduled pipeline, instructions are issued out of order, executed out of order, completed out of order and committed in order.

False. In a dynamically scheduled pipeline, instructions are always issued in program order, but they may execute and complete out of order. They are committed/retired in program order. (1+1)

(e) If the 32-bit representation of two IEEE-754 floating point numbers are 0x41C80000 and 0x41820110, respectively, then their absolute difference is more than 16.

False,

A=0x41C80000

Conversion to binary,

0 100 00011100 1000 0000 0000 0000 0000

E' = 100 00011 = 131

E=131-127=4

+ 1.1001000x2^4 = 11001.000 = +25

B= 0x41820110

Conversion to binary,

0100 00011000 0010 0000 0001 0001 0000

E'= 100 00011= 131

E=131-127= 4

+1.00000100000000100010000 x 2^4 = 10000.010000000100010000=

+16.25 (approx)

Absolutive difference = 8.75 (approx). Their absolute difference is not more than 16. (1+1)

(OR)

Both values have the same exponent (2^4).

Only their mantissa values differ. So their value will be between 16 to 32.

Hence their absolute difference can’t be more than 16.

CS 2201 – Computer Architecture & Organization – Part B

Name:	Roll Number:	Invigilator Sign:
Marks: (For Examiners Use)	Q9:	Q10:

9. A program has a for loop that runs for 8 iterations and hence there are 8 branch instructions to be executed. Every alternate iteration is taken, starting from the first one as taken. Hence there are total of 4 taken branches and four not taken branches alternatively in the sequence. This program is executed on a processor that uses a standard (2, 2) correlating branch predictor with initial BHT values as 00, 00, 11, and 11 for NN, NT, TN, and TT, respectively. Illustrate the BHT changes for the 8 iterations of the branch by filling up the following table and answer the question given below.

S. No	Last Outcome	BHT (NN/NT/TN/TT)	Prediction	Outcome	Remarks
1	NN	00 / 00 / 11 / 11	N	T	misprediction
2	NT	01 / 00 / 11 / 11	N	N	Correct
3	TN	01 / 00 / 11 / 11	T	T	Correct
4	NT	01 / 00 / 11 / 11	N	N	Correct
5	TN	01 / 00 / 11 / 11	T	T	Correct
6	NT	01 / 00 / 11 / 11	N	N	Correct
7	TN	01 / 00 / 11 / 11	T	T	Correct
8	NT	01 / 00 / 11 / 11	N	N	Correct

[2 Marks]

(a) What is the final state of BHT after execution of the 8th iteration?

Ans: 01 / 00 / 11 / 11 [2 Marks]

(b) What is the accuracy of the branch predictor?

Ans: ⅞ x 100 = 87.5% [2 Marks]

(c) What initial value of BHT could have given 100% accuracy in branch prediction for this loop? Explain.

Ans: 1x/0x/11/11 [2 Marks]

If the BHT entry is not completely written in full and only "TN" is written, -1 mark.

…… ***Calculations, Illustration and Rough workspace for Q.No:9*** …….

10. Consider a multicycle speculative instruction pipeline with an issue width of 1 and ROB size of 4 that uses Tomasulo's algorithm with two reservation stations per functional unit. With two reservation stations per unit, operation of one can be in execution stage and the other can be in the reservation station buffer waiting for its turn. There is one 7-cycle FMUL unit and one 4-cycle FADD unit, all connected to a single CDB. Assume that both functional units are internally pipelined. An instruction waiting for data on CDB can move to its EX-start in the cycle after the CDB-write. If an instruction issue is stalled due to ROB structural hazard, then the stalled instruction moves to EX-start after once cycle of the ROB head entry commit. The following 5 instructions are to be issued. Each instruction has 2 source operands (Op1 and Op2). In the table, Op1 and Op2 should be filled at time of issue with either RF / ROB / CDB depending on from where the respective operand has to be fetched.

I No	Instruction	Op1	Op2	Issue	EX-start	CDB-write	Commit
I1	FMUL F1, F2, F3	RF	RF	1	2	9	10
I2	FADD F3, F1, F2	CDB	RF	2	10	14	15
I3	FADD F4, F5, F6	RF	RF	3	4	8	16
I4	FMUL F1, F5, F4	RF	CDB	4	9	16	17
I5	FMUL F5, F2, F4	RF	ROB	10 / 11	11/ 12	18 / 19	19 / 20
I6	FADD F6, F4, F5	ROB	DCB	15 / 16	19 / 20	23 / 24	24 / 25

[1 Mark for each row. At least one entry is wrong, 0 mark]

(a) What will be the value changes of RSI (F1) from clock cycle 1 to clock cycle 25?

Ans: C1-C3 => ROB(I1)

C4-17 =>ROB(I4) [1 Mark for C1-C3 and C4-C17]

C8-C25 => 0 [1 Mark for C18-25, if it's not written, 0 MARKS]

(b) What is the CPI of execution of this program?

Ans: 24/6 or 25/6 [1 Mark, if the table is also correct]

…… ***Calculations, Illustration and Rough workspace for Q.No:10*** ……